MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

HORTICULTURE SCIENCE PAPER-I

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. Which plants hormone plays an important role in apical dominance?

(a) Gibberellins (b) Auxin

(c) ABA (d) Cytokinins

2. Abundant flowering and fruiting with lessened vegetative growth can result in which condition?

(a) High 'C' and relatively lower 'N' content (b) High 'N' and relatively lower 'C' content

(c) Low 'N' and low 'C' content (d) High 'C' and high 'N' content

3. Dioecious form of sex is found in:

(a) Citrus (b) Aonla

(c) Papaya (d) Litchi

4. Which is a climacteric type of fruit?

(a) Grape (b) Orange

(c) Litchi (d) Papaya

5. Study for production and cultivation of grapes is known as:

(a) Floriculture (b) Arboriculture

(c) Viticulture (d) Olericulture

6. Herbaceous stem cutting is popularly done in:

(a) Chrysanthemum (b) Nerium

(c) Emblica (d) Lotus

7. Heterostyly condition is observed in:

(a) Tomato (b) Brinjal

(c) Potato (d) Onions

8. *In-situ* grafting is practiced in:

(a) Laboratory (b) Planting site

(c) Test tube (d) Aerial parts

9. Mevalonic acid is a precursor of:

(a) IAA (b) NAA

(c) ABA (d) BAP

10. Rossette formation is seen due to deficiency of:

(a) Mn (b) Zn

(c) Cu (d) Mo

11. Epigeal germination is seen in:

(a) Cashew (b) Coconut

(c) Mango (d) Citrus

12. The ratio of NPK in starter solution should be:

(a) 2:1:1 (b) 2:2:1

(c) 1:2:1 (d) 1:2:2

13. Which crop is the most susceptible to water logging:

(a) Banana (b) Pomegranate

(c) Papaya (d) Jack fruit

14. 'Breeder seed' is the source of :

(a) Certified seeds (b) Nucleus seeds

(c) Foundation seeds (d) Registered seeds

15. For breaking bud dormancy, suitable chemical is-

(a) Abscisic acid (b) NAA

(c) Ethylene (d) IBA

16. Which one is free-living nitrogen fixing biofertilizer?

(a) Rhizobium (b) Azotobacter

(c) Pseudomonas (d) Penicillium

17. Which PGR is used to induce femaleness in fruit?

(a) GA (b) ABA

(c) 2,4-D (d) Maleic Hydrazide

18. In biodynamic farming, known formulation BD-501 is made with:

(a) Cow horn silica
(b) Chamomile in cow intestine
(c) Himalayan stinging nettle
(d) Himalayan oak bark

19. Which method of grafting is most used in temperate fruit propagation?

(a) Epicotyl grafting
(b) Wedge grafting
(c) Tongue grafting
(d) Approach grafting

20. Most commonly used training system in Kiwi fruit production:

(a) Espalier
(b) Spindle bush
(c) Modified Central Leader
(d) T-bar

21. Which micronutrients are commonly used for controlling mango fruit drop?

(a) Co &Ni
(b) B &Zn
(c) Ca &Fe
(d) S &Mg

22. As the soil pH increases, the relative proportion of which ion decreases?

(a) $H_2PO_4^-$
(b) HPO_4^{2-}
(c) PO_4^{3-}
(d) HPO_4^-

23. At field capacity the value of pF is:

(a) 0.0
(b) 2.5
(c) 4.2
(d) 6.0

24. The critical stage for irrigation in garden pea is:

(a) Sowing
(b) Leaf initiation
(c) Pod setting
(d) Harvesting

25. Botanical fruit type of pomegranate is:

(a) Drupe
(b) Berry
(c) Sorosis
(d) Balausta

26. Chemical formula of 'Dolomite', commonly used chemical in correction of soil pH is:

(a) $CaCO_3$
(b) $MgSO_4$
(c) $Ca(OH)_2$
(d) $CaMg(CO_3)_2$

27. Per cent content of sulphur in single super phosphate is:

(a) 8
(b) 12
(c) 16
(d) 20

28. Which one is not a type of greenhouse design?

(a) Quonset
(b) Gothic arch
(c) Saw-tooth
(d) K-frame

29. Which one is salt resistant rootstock of grapes?

(a) St. George
(b) River Bank
(c) Dogridge
(d) Saltdot

30. The effect of 'Vernalization' on flower induction can be reversed by treatment with:

(a) light exposure (b) blocking lights

(c) low temperature (d) high temperature

31. In Hasth Bahar, fruit ripening of guava occurs in-

(a) February-April (b) October-November

(c) July-September (d) November-January

32. 'Cauliflory' is a specialized bearing habit observed in which fruit crop?

(a) Mango (b) Ber

(c) Sapota (d) Jackfruit

33. Which *Psidium* species is a nematode resistant rootstock?

(a) *Psidium cujavillis* (b) *Psidium friedrichsthalianum*

(c) *Psidium pumilum*

34. Which one is a non-selective herbicide?

(a) Atrazine (b) Fluchloralin

(c) Paraquat (d) Butachlor

35. Which endogenous hormone is responsible for seed dormancy?

(a) IAA (b) ABA

(c) Ethylene (d) IBA

36. Which one is a polyembryonic mango variety?

(a) Tomy Atkins (b) Mehmood Behar

(c) Olour (d) Lakhsman Bhog

37. Which one is an important trap crop for nematode?

(a) Marigold (b) Chrysanthemum

(c) Hibiscus (d) Aster

38. Full form of IFOAM is:

(a) International Foundation of Organic Agriculture Movements

(b) International Federation of Organic Agriculture Movements

(c) Indian Federation of Organic Agriculture Movements

(d) International Foundation of Organic Agriculture Measures

39. Which one is not used as glazing material in green house construction?

(a) Glass

(b) Rubber wood

(c) Fibre glass reinforced plastic (FRP)

(d) UV stabilized low density polyethylene (LDPE)

40. *Trichogramma brasiliensis* work as egg parasite to control which pest?

(a) Fruit borer in tomato (b) Epilachna beetle on potatoes

(c) Chilli thrips (d) Nematode in brinjal

41. Which fruit does not have double sigmoid growth curve?

(a) Grape (b) Mango

(c) Peach (d) Persimmon

42. Multi-tier cropping is commonly practiced in which horticulture crop?

(a) Grape (b) Fig

(c) Guava (d) Coconut

43. Which method of grafting is followed for topworking while rejuvenating old mango orchard?

(a) Tongue grafting (b) Whip grafting

(c) Cleft grafting (d) Epicotyl grafting

44. Which one is most potent ethylene inhibitor?

(a) ABA (b) 1-MCP

(c) GA (d) Brassinolide

45. Among these, which micro-irrigation method is generally having highest discharge rate?

(a) Drip (b) Sprinkler

(c) Bubbler (d) Pitcher

46. Whiptail of cauliflower is due to which deficiency?

(a) B (b) K

(c) Mo (d) Ca

47. Lycopene is found in high quantity in which vegetable crop?

(a) Radish (b) Parsley

(c) Tomato (d) Pumpkin

48. Panicle (Spadix) type inflorescence found in which crop?

(a) Tamarind (b) Coconut

(c) Cocoa (d) Sapota

49. Which one is not animal based concentrated organic manure?

(a) FYM (b) Bone Meal

(c) Horn Meal (d) Hoof meal

50. Which method of irrigation is mostly followed in fruit tree cultivation?

(a) Furrow (b) Check basin

(c) Ring Basin (d) Flood

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any 5 (five)** questions.*

1. Describe briefly the factors responsible for fruit drop. How do you control fruit drop problem? Explain the objectives and methods for fruit thinning. **(6+4+5+5=20)**

2. What are plant bio-regulations? Briefly describe their functions. What are the different types of auxins? Briefly write about their functions and applications. **(2+8+3+7=20)**

3. Give short note on ***any four***: **(4×5=20)**

(a) Vermicompost
(b) Crop rotation
(c) Biofertilizer
(d) Green manure
(e) Concentrated Organic manure

4. What is budding? Describe briefly about different techniques of budding with suitable diagram. Describe how rootstock can influence the performance of scion cultivar in a stock-scion relationship. **(2+8+10=20)**

5. Describe with layout diagram different surface irrigation systems. What are the advantages of micro-irrigation over conventional irrigation? Give a brief description of different micro-irrigation methods. **(10+4+6=20)**

6. What are the essential nutrients for plants? Make a list of it and explain the difference between macronutrients and micro nutrients. Give example of two macro and two micronutrients with their role and deficiency symptoms. **(2+6+12=20)**

7. Define Integrated Nutrient Management. What are the prime objectives of INM? Briefly write about the methods of INM. Write in short about advantages and limitations of INM. **(2+4+8+6=20)**

8. What do you understant by weed? Explain the concept of Integrated Weed Management (IWM). What is the need for IWM? Give a brief account of different methods for weed control. **(2+4+4+10=20)**

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